

Rate of turn, turn radius and the steering linkage — Northstar 76, 2026 season

26 September 2026. Every number is measured from the boat's own logs. Where the logs cannot support a measurement it is left out rather than estimated, and the coverage is stated beside each figure.

1. The three numbers, and why they differ

Rate of turn — heading change over the **6 seconds centred on the moment the apparent wind crosses the bow**, divided by the span actually measured. The average through the fastest part of the turn.

Turn radius — **speed ÷ yaw rate** over that same window. The radius the boat actually turned at. This is the operating point any steering geometry has to be correct for, and unlike the geometry it can be measured without knowing anything about the boat.

Max rotation — the single fastest heading change between two consecutive samples. A *peak*, which rises with the log's own sampling noise.

Compared **on the same manoeuvres**, max rotation reads higher in **94 of 98 tacks** and **44 of 44 gybes**, by a median of **1.00 °/s** and **1.19 °/s**. That gap is the sampling, not the sailing. A target set from max rotation would be a target nobody is actually sailing.

Timing the window on the apparent wind rather than on the manoeuvre detector matters: on 12 September the crossing lands a consistent **+2.7 s after** the detected moment, so timing from the detector would have measured a different part of every turn.

What the season can carry

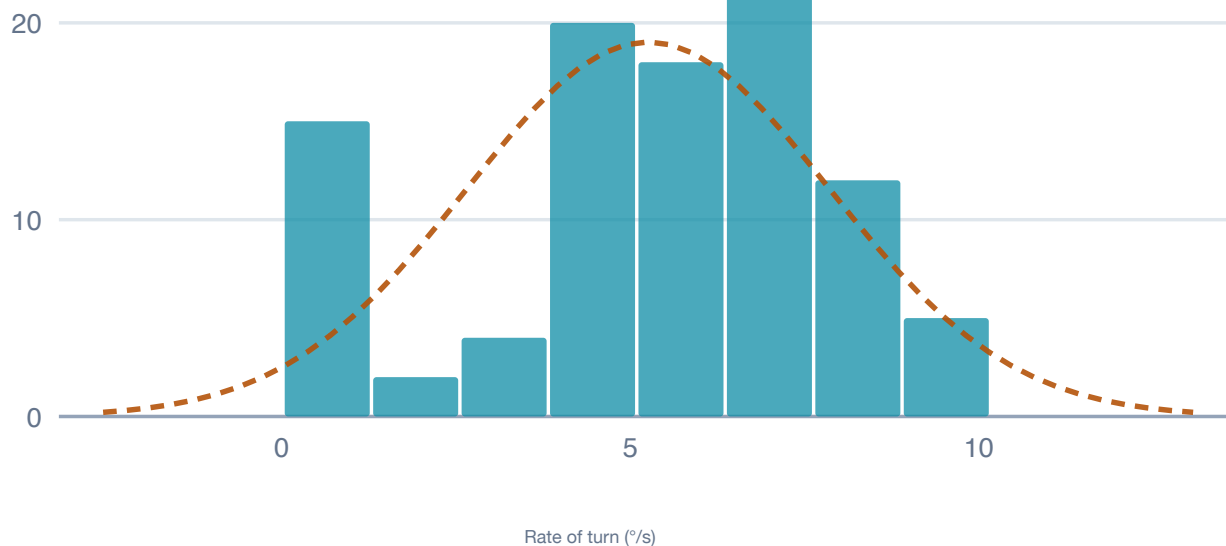
	tacks	gybes
Rate of turn (±3 s window)	98 of 587, on 8 of 23 days	44 of 230, on 6 of 22 days
Turn radius	81	42
Max rotation	546	220

Rate of turn needs a log sampled at **3 s or finer**; half the season is logged at 6.09 s, and a six-second window sampled every six seconds is not a measurement. Turn radius needs the rate *and* a real turn under it — **$R = v/\omega$ runs away as ω goes to zero**, so it is only computed above 2 °/s. Max rotation survives a coarse log, which is why it covers five times as many manoeuvres, and why it should be read as a ceiling rather than a performance.

2. Tacks

Rate of turn across 98 tacks

■ Rate of turn · 98



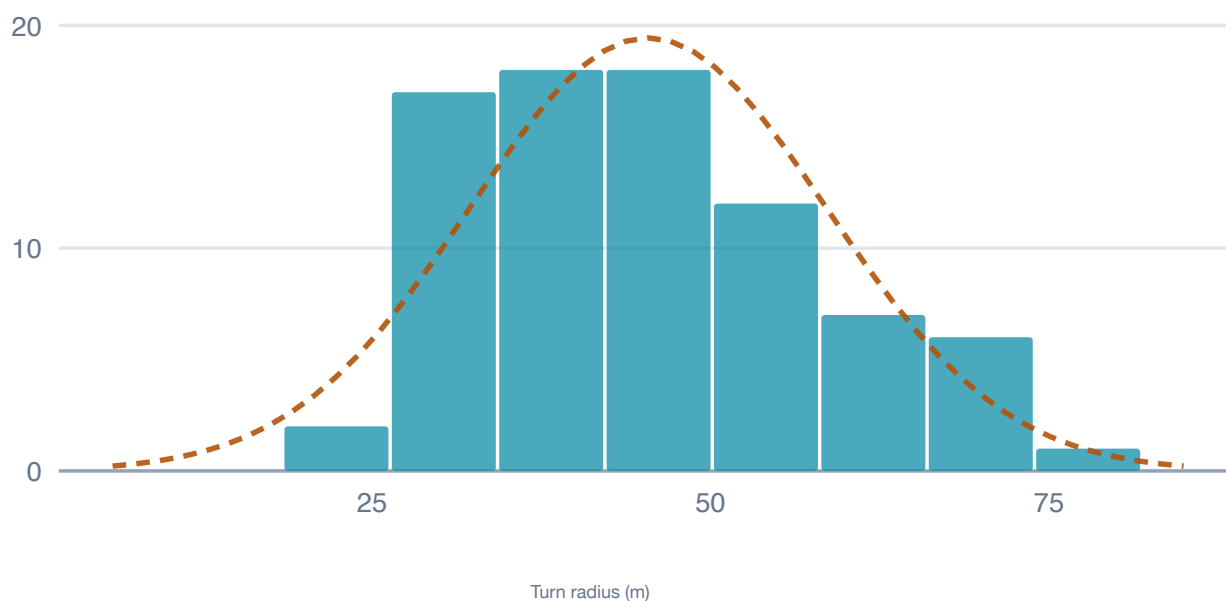
Rate of turn through 98 tacks. Bars are counts; the dashed line is the normal fitted to the same mean and spread, drawn so the difference from it can be seen.

It is not a bell. A hump at 5–7 °/s and a tail of slow ones running to nearly zero. The mean of **5.25 °/s** sits below the median of **5.73** because that tail drags it there; the typical tack is the median.

The bottom bar is not slow tacking. **Fifteen of the ninety-eight** sit near zero, and a tack that turns at 0.3 °/s is something recorded as a tack that was not one, or a stretch where the heading data is bad. They are excluded from the radius for that reason.

Turn radius across 81 tacks

■ Turn radius · 81

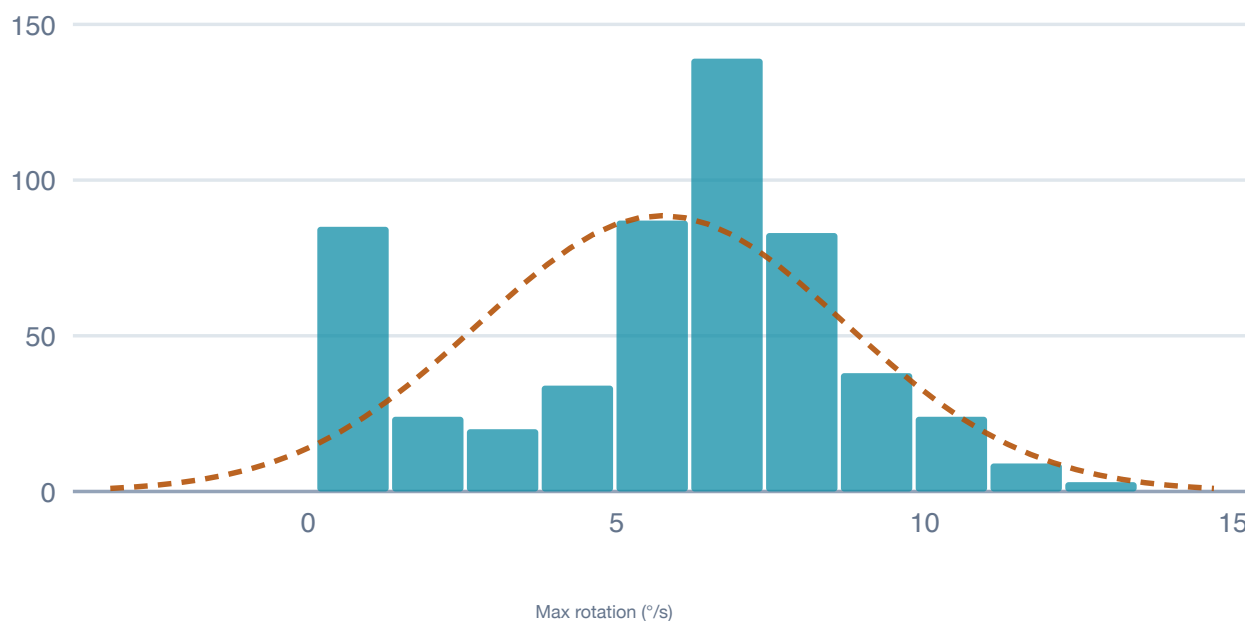


Turn radius through 81 tacks, in metres.

The boat tacks in about 43 m — median, against a hull length of 23.16 m, so **1.86 boat lengths**, and tightly grouped: 18 m to 82 m with a spread of 13 m.

Max rotation across 546 tacks

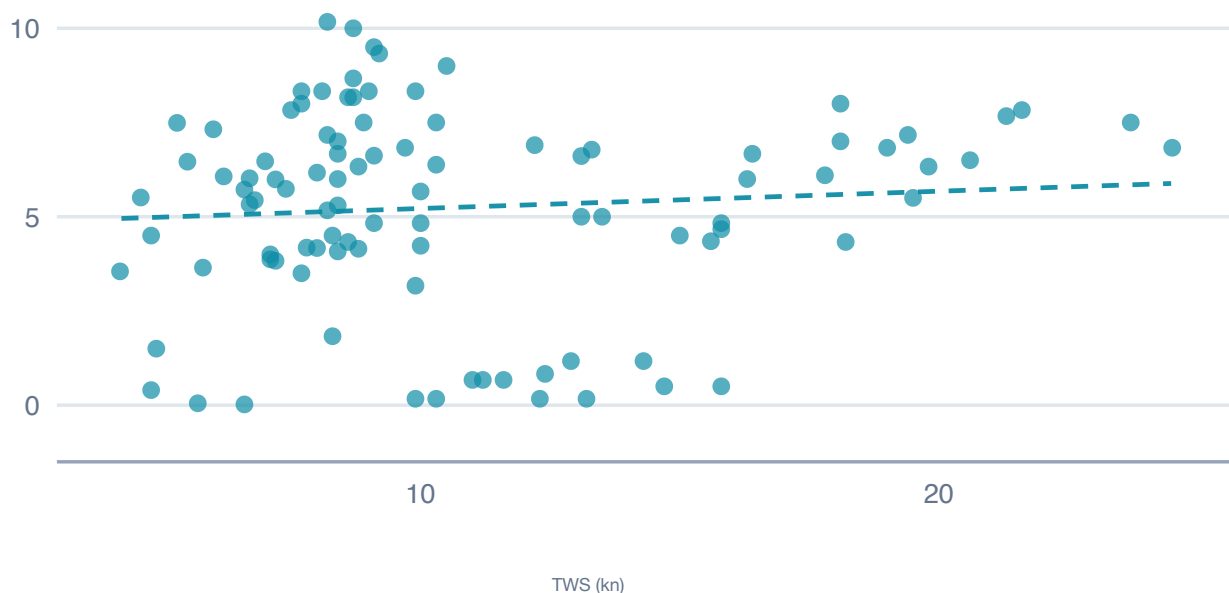
■ Max rotation · 546



Max rotation through 546 tacks — five times the sample, because a coarse log still supports it.

Rate of turn vs TWS (°/s)

● Tacks · 98 · R^2 0.01

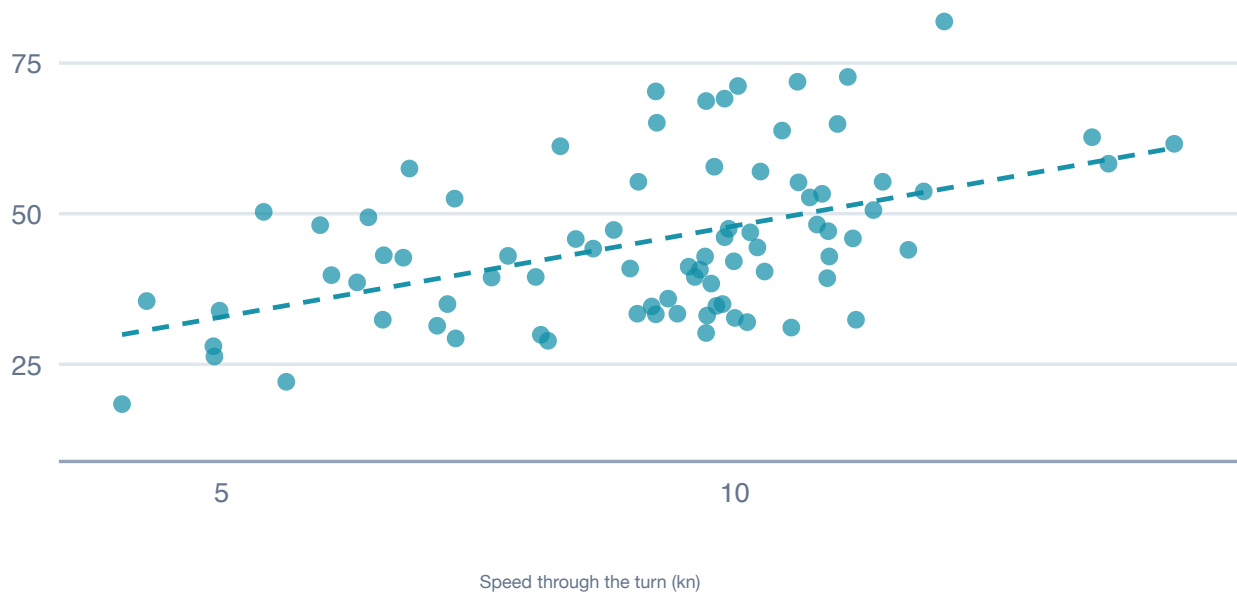


Rate of turn against true wind speed, 98 tacks, with a least-squares line.

Rate of turn does not depend on the breeze. R^2 is **0.01** across 7–20 knots. Whatever governs how fast this boat comes through the wind, it is not wind strength — so it is worth looking at what the crew does rather than at conditions.

Turn radius vs Speed through the turn (m)

● Tacks · 81 · R^2 0.24

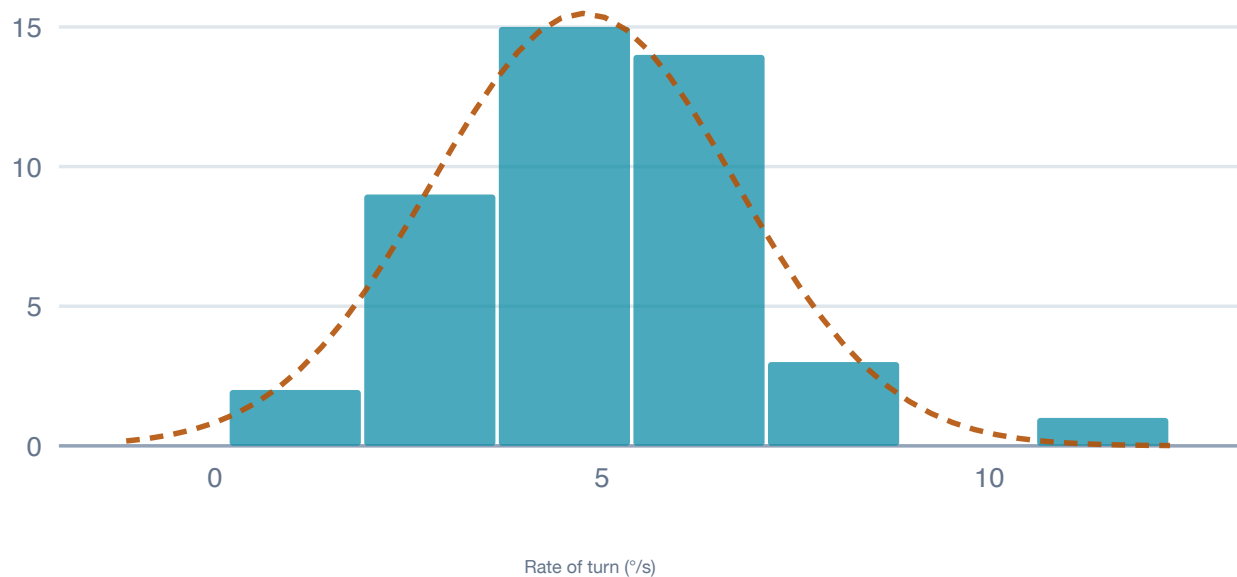


Turn radius against the speed carried through the turn.

3. Gybes

Rate of turn across 44 gybes

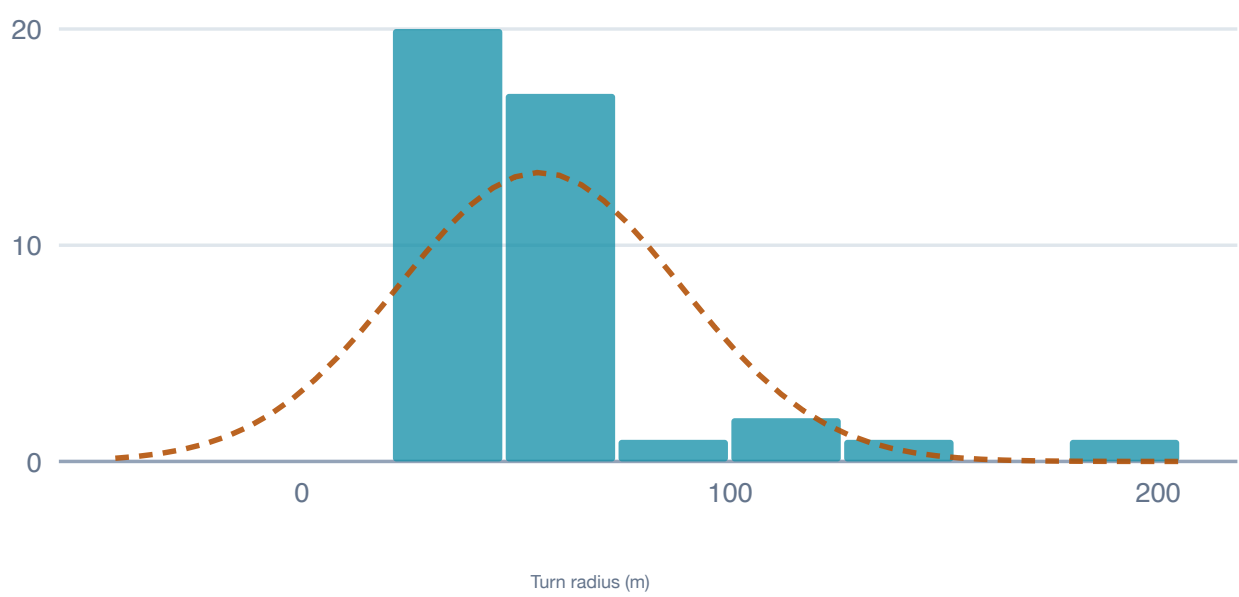
■ Rate of turn · 44



Rate of turn through 44 gybes.

Turn radius across 42 gybes

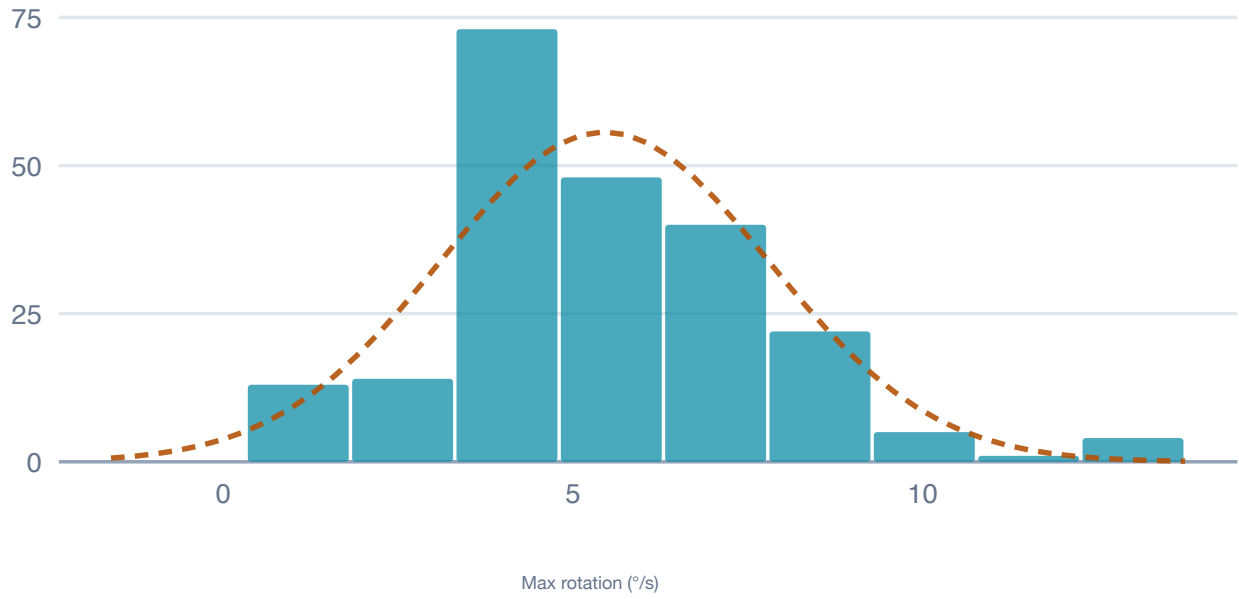
■ Turn radius · 42



Turn radius through 42 gybes.

Max rotation across 220 gybes

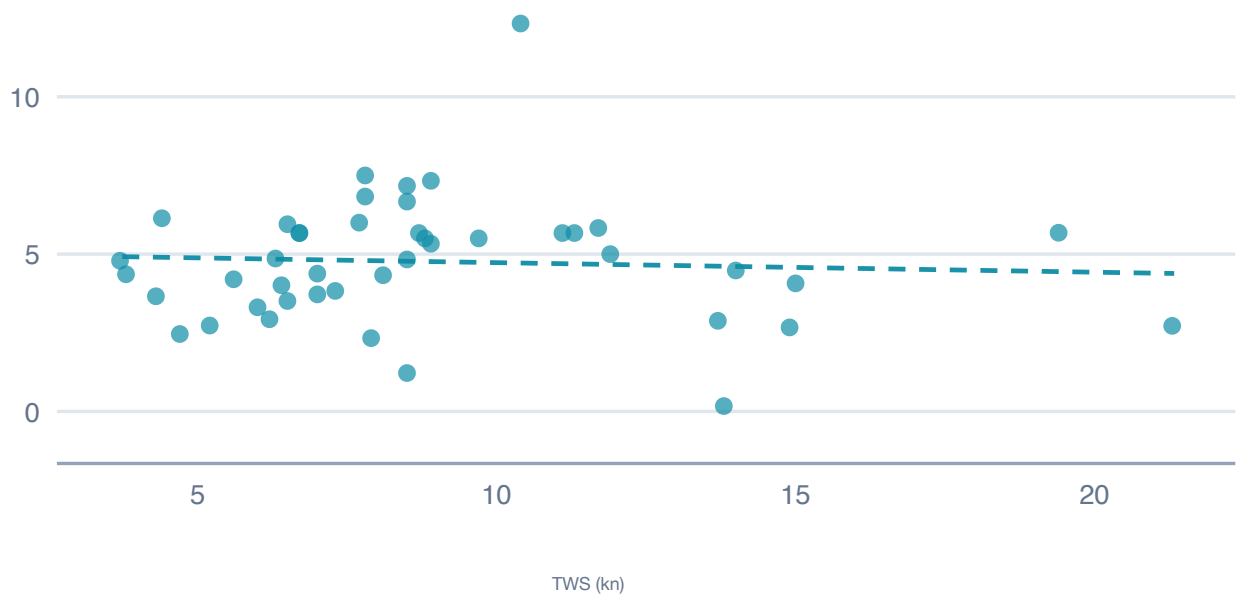
■ Max rotation · 220



Max rotation through 220 gybes.

Rate of turn vs TWS (°/s)

● Gybes · 44 · R² 0.00



Rate of turn against true wind speed, 44 gybes.

Gybes turn **slower and wider** than tacks — 4.81 °/s against 5.73, and a radius of **49.6 m (2.15 boat lengths)** against 43.0 — but the rate is more tightly grouped (spread 1.97 against 2.61). A gybe is a smaller heading change made under more control, and it is the more repeatable manoeuvre on this boat. The radius spread is the wider of the two (33 m against 13 m), driven by a handful of very wide ones out to 205 m.

4. Summary

- The typical tack turns at **5.7 °/s** in **43 m**; the typical gybe at **4.8 °/s** in **50 m**. Medians, from the ± 3 s window.
- The boat turns in **roughly two boat lengths**, and tacks are the tighter of the two manoeuvres.
- The tack rate distribution is **not normal** — a hump with a tail of slow ones. The tail is where the time goes, not the average.
- **Rate of turn is independent of wind speed** (R^2 0.01). If it is to be improved, the variable is not the breeze.
- **Fifteen of the ninety-eight rated tacks look like bad records rather than bad tacks** and should be checked before any mean is quoted onward.
- Max rotation reads higher on the same manoeuvre almost every time. Use it for coverage, not as a target.

5. Steering geometry

The operating point, measured

This is what turn radius buys: the geometry only has to be right where the boat actually sails, and that is now known rather than assumed.

Treating the boat as a bicycle — a single effective steering angle at an effective distance **L** ahead of the turn centre — the helm carried through each turn gives

$$L = R \cdot \tan \delta$$

Across **68 tacks** that lands at a median **12.6 m** (range 5.5–19.6), and across **36 gybes** at **11.7 m**. The two agree, which they need not have, and both are close to **half the hull length**.

That is an *effective* pivot distance, not a tape measurement: a hull generates side force along its whole length, so it will not equal the geometric distance from the rudder stock to the centre of lateral resistance. It is, however, the number the boat behaves as though it has — **worth comparing against the tape measurement rather than replaced by it**.

At the measured operating point — radius **43 m**, L **12.6 m** — the helm angle implied is **16.3°**, which matches the top of the measured steering range.

The linkage, measured

Across **78,149 log rows** carrying both rudder angles, fitting starboard against port gives

$$\text{ruddS} = 1.0008 \times \text{ruddP} - 1.215 \quad (R^2 \ 0.932)$$

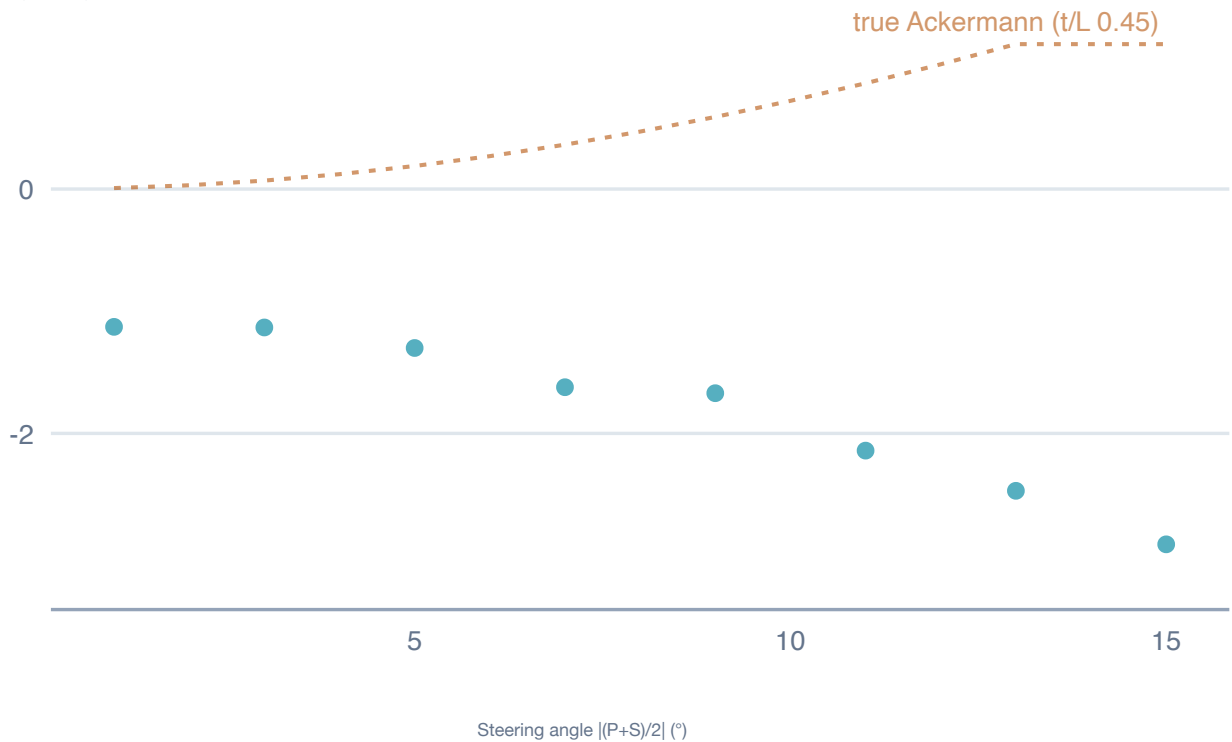
A slope of **1.0008** is **parallel steering**: within these sensors' resolution the two rudders turn by the same amount, with a **constant offset of about –1.2°** rather than a progressive differential.

The differential does drift with steering angle — weakly, and the *wrong way*:

steering angle	rows	mean starboard – port
0–2°	43,058	-1.13°
2–4°	19,228	-1.13°
4–6°	6,996	-1.30°
6–8°	3,633	-1.62°
8–10°	1,990	-1.67°
10–12°	1,359	-2.14°
12–14°	1,042	-2.47°
14–16°	842	-2.91°

Rudder differential against steering angle (°)

● Measured (binned) · 78149



Measured differential (binned means) against steering angle, with the differential true Ackermann would require shown dashed. The measured curve runs the other way.

About **-0.09 to -0.15° per degree** of helm depending on turn direction, with R^2 of only **0.05 to 0.14** — a small systematic effect inside a lot of scatter.

Under true Ackermann the inner rudder turns **more** than the outer, making starboard – port **positive** in both directions. It is negative and grows more negative. **The boat is running slightly anti-Ackermann.**

The target, once the rudder separation is known

With **L = 12.6 m measured** and the operating radius of **43 m**, the only missing number is **t**, the separation between the rudder stocks. Inner and outer angles at that operating point:

t (m)	t / L	inner	outer	differential needed
4.0	0.317	17.08°	15.64°	+1.44°
4.5	0.357	17.18°	15.56°	+1.62°
5.0	0.397	17.28°	15.48°	+1.80°
5.5	0.437	17.38°	15.40°	+1.98°
6.0	0.476	17.48°	15.32°	+2.17°

Against a **measured -2.91°** at that helm. Whatever **t** turns out to be, the correction is of the order of **4.4° to 5.1° at full helm** — reversing the sign and adding the Ackermann. That is large enough to be worth getting right, and large enough that guessing **t** would have been irresponsible.

The one thing that must happen first

These are sensor readings, not linkage geometry. A constant **-1.2°** offset is indistinguishable, from the log alone, from one rudder's angle sensor being zeroed **1.2°** differently from the other. **Zero both sensors dockside against physically aligned rudders and record the offsets before anything mechanical is touched.** If the offset proves to be calibration, what remains is the **~0.1 °/°** drift, which is small — and the whole picture changes.

The linkage: what to actually build, for t = 4.0 m

The rudder separation is fixed at **4.0 m** and the adjustable geometry is the tiller arm. Solving the steering trapezoid — two tiller arms joined by a tie rod — against the measured operating point gives a definite answer, and one correction to the premise.

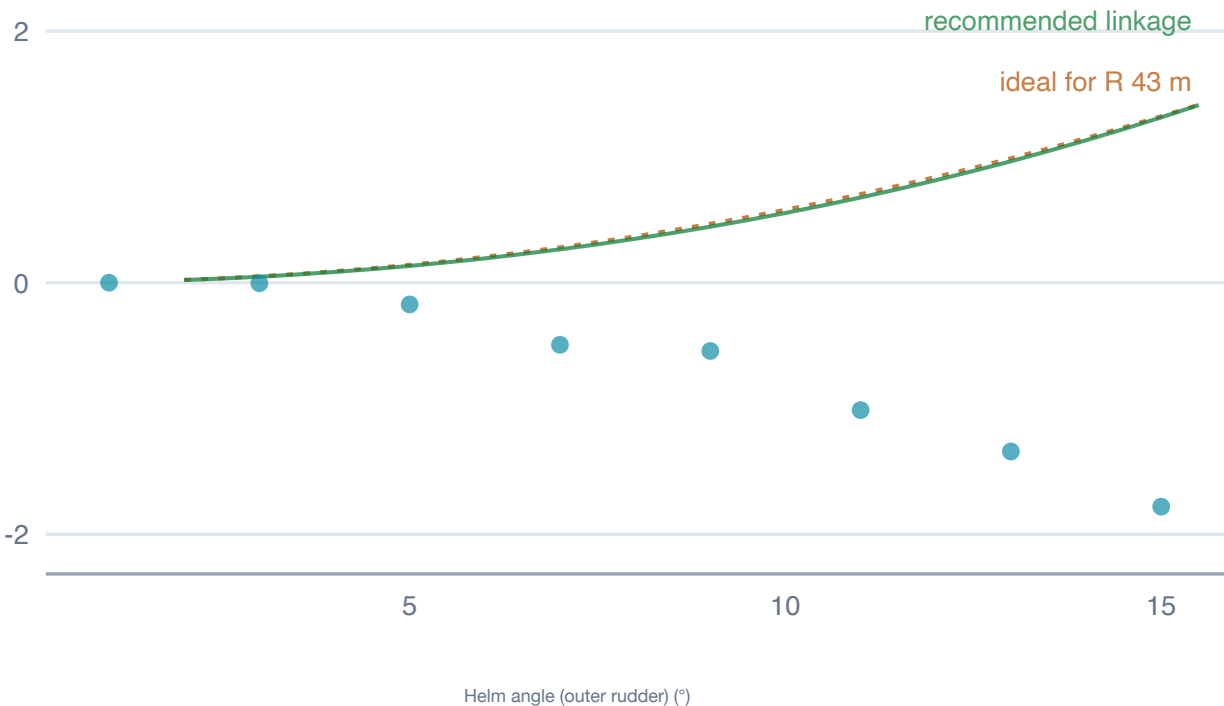
The tiller LENGTH is not the lever. The arm ANGLE is. Stretching the tiller from 200 mm to 1000 mm — a five-fold change — moves the differential by only **+9 % to +42 %**, depending on how far the arms are already angled. It cannot carry the boat from anti-Ackermann to correct Ackermann. What sets the Ackermann is how far **inboard** the tie-rod ball joint sits relative to the rudder stock.

tiller length	required arm angle	ball joint inboard	tie rod
300 mm	15.9°	82 mm	3836 mm
400 mm	15.7°	108 mm	3784 mm
500 mm	15.4°	133 mm	3734 mm
600 mm	15.2°	158 mm	3685 mm

The angle barely moves with length, which is the same fact seen from the other side: **put the ball joint on a line running 15.5°–16° inboard of the fore-and-aft line from each rudder stock**, at whatever radius the hardware allows, and shorten the tie rod to suit. In practice that is a new hole, not a new arm.

Rudder differential: measured, recommended, ideal (°)

● Measured now (offset removed) · 78149



What the recommended linkage delivers across the steering range, against the ideal for a 43 m turn, with the measured behaviour (constant offset removed) for comparison. The solid line is the linkage; it tracks the ideal to within 0.1° out to 15° of helm.

Across the working range that geometry tracks true Ackermann closely:

helm	recommended	ideal
5°	+0.13°	+0.14°
10°	+0.55°	+0.58°
15°	+1.31°	+1.32°
20°	+2.50°	+2.37°

It runs slightly over Ackermann past 20° , which is normal for a trapezoid linkage and harmless — the boat does not sail there.

One number that does not add up, and what it means

Working backwards from the measured behaviour, a symmetric tie-rod linkage would need its arms splayed **about 28° outboard** to produce the drift seen. That is a great deal of splay — enough to doubt the premise rather than the measurement.

Taken with the constant -1.2° offset, which is indistinguishable from one rudder sensor being zeroed differently, the likeliest reading is that **part of what looks like anti-Ackermann is not geometry at all**: sensor zero, sensor non-linearity, or flex in the loaded rudder under helm. The R^2 of 0.05–0.14 on that drift says the same thing.

So the table above is what to build **once the linkage has been looked at**, not before.

In order

1. **Zero both rudder sensors** dockside against physically aligned rudders, and record the offsets. This alone may account for the -1.2° .
2. **Measure the tiller arms as they are** — length, and how far inboard or outboard the tie-rod ball joint sits from the stock. If they are near parallel, the drift is instrumentation and the linkage is simply zero-Ackermann. If they really are splayed outboard, that is the fault and the table above is the fix.
3. **Confirm L.** The 12.6 m is what the boat behaves as though it has; compare it against the tape measurement from the rudder stock to the centre of lateral resistance. If the tape says something very different, the target differential moves and the table should be recomputed.
4. **Get more days onto a 1–2 s log.** Rate of turn and radius exist on 8 days of 23 only because the rest are logged at 6.09 s — and it is the radius that sets the whole geometry.
5. **Then** move the ball joint to the line above, shorten the tie rod, and re-measure the differential against this same chart.

Generated from `session_phase_stats` and the session logs, 26 September 2026. Linkage solved numerically as a symmetric steering trapezoid (two tiller arms, one tie rod, rudder separation 4.0 m), verified against the parallel-arm case which must and does return exactly parallel steering. Rate of turn: heading change over the 6 s centred on apparent wind = 0, on logs of 3 s or finer, gated on the crossing being interpolated across ≤ 3 s and the window holding ≥ 3 headings. Turn radius: mean boat speed over the same window divided by the yaw rate, above $2^\circ/\text{s}$ only. Linkage fit: ordinary least squares over log rows carrying both rudder angles with $|\text{steering}| \leq 15^\circ$. Effective pivot distance: $R \cdot \tan \delta$ with δ the mean helm over the turn, 2° – 25° .